A PILOT STUDY ON USING PROBIOTICS TO REDUCE THE APPLICATION RATE OF NITROGEN FERTILIZER BASED ON ALTERNATE WETTING AND DRYING (AWD) IRRIGATION

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ABSTRACT

Alternate wetting and drying (AWD) irrigation is an optimized agricultural water management practice with much less water than the usual system of continuously 'flooding' the field for planting rice. Long-term fertilizer application can cause the soil acidification and result in poor crop growth and significantly greater yield reductions. To reduce the application rate of fertilizer based on the AWD irrigation, the performance of probiotics that incorporated with the reduced fertilizer were investigated on the quantitative traits. The results indicated that the nitrogen fertilization rate could be reduced. Taking plant height as an example, there is no significant difference among the 25, 50 75 and 100% fertilization rates during the vegetative period, except the non-fertilized treatment. Moreover, the performance of rice growth status was consistent expressed in SPAD value among the 25, 50 75 and 100% fertilization rates. However, the performance of tillers number values is similar within fertilized groups, and is significantly higher than non-fertilized group. These methods reduce water and nitrogen fertilizer demand without reducing crop yields. In this study, plant height in fertilized groups showed two stunted interval, that are mostly related to maximum tillering and panicle initiation stages, but in the non-fertilized group it gently increased. SPAD values gradually decreased in fertilized groups, and were significantly higher than non-fertilized group upto 50 days. NDVI values showed complicated trends in all treatments, that means the interaction between beneficial microorganism and AWD irrigation approach probably existed. Accurate assessment of rice growth conditions in the entire season, especially the tipping point in panicle initiation stage, which affects yield potential, is essential for farmers and researchers to optimize field practices and further investigation is required for future industrial applications.

Keywords: alternate wetting and drying (AWD), probiotics, nitrogen fertilizer

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1. INTRODUCTION

Water scarcity which is predicted to get worse is a serious problem in agriculture that should be faced in many parts of the world (Schewe et al., 2014). The serious regional and seasonal water shortage, and the increased demand for water increment in city and industries have further intensified water scarcity. Irrigation of agricultural lands is the largest consumer of water at global level that accounts for 70% of the water used worldwide. Moreover, 75% of total rice production over the world comes from irrigated lowlands (Guerra et al., 1998). On the other hand, rice is one of the most important crops in the world. During rice planting, flooded rice requires two to three times more water than other cereal crops such as wheat and maize (Bouman et

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al., 2007). Furthermore, groundwater is one of the major water resources for agriculture of the world, even in Southern Taiwan. Excessive groundwater pumping will increase the problem of water shortage. Therefore, water resource is an important limiting factor in rice planting.

Recently, various water saving techniques have been developed to improve the water use efficiency for rice producers with the advantage of decreasing water demand and maintaining acceptable yield. One of the most widely promoted water-saving technique for rice is alternate wetting and drying (AWD) technique. The AWD irrigation approach was introduced to sustain the rice production system and food security. The principle of this method is to irrigate the soil alternately between flooding water and drying until cracking soil conditions. Compared with traditional continuously flooding irrigation, rice yield can be maintained or even increased slightly in systems of AWD, while water inputs were reduced by around 15-30% (Tan et al., 2013). However, due to the factor that the nitrification-denitrification processes induced by drying and wetting cycles, AWD rice systems have greater potential on nitrogen losses compared to conventional flooding method (Broadbent, 1971; Sah and Mikkelsen, 1983; Mikha et al., 2005; Borken and Matzner, 2009; Tan et al., 2013). Therefore, in our study, nitrogen fixation probiotics was tried to integrate with AWD rice systems. Plant probiotic bacteria may directly or indirectly exert beneficial effect on plant growth and development by the free-living plant-associated bacteria (Glick, 2012; Islam and Hossain, 2012; Khan et al., 2017). By the application of nitrogen fixation probiotic, the nitrogen losses could be offset that result in maintaining the rice growth and development. Furthermore, it might advantage to decrease the application of extra chemical fertilizer and the use of water. Therefore, the main objective of this study is to assess the efficiency of the integrated AWD with probiotics application, by reducing the water and nitrogen fertilizer usage, providing an alternative approach to achieve eco-friendly production of paddy rice.

2. MATERIALS AND METHODS

2.1 Experimental Site and Trial Design

To investigate the performance of probiotics incorporated with the different fertilization rate on the quantitative traits on rice growth and development based on the AWD techniques, the experiments were conducted in 2018 at the experimental field (22°38' N, 120°36' E) of National Pingtung University of Science and Technology, Pingtung, Taiwan. A semi-dwarf japonica cultivar, Kaohsiung147 (KH147) developed by Kaohsiung District Agricultural Research and Extension Station, was selected as target crop.

The experiments were designed as completely randomized block design with three replications and five fertilization rates. For each replication, three seedling were planted in12 inches cultivation pot. The full amount of nitrogen fertilizer was 160 kg/ha in conventional farming in Taiwan. Five fertilization rates were investigated incorporated with probiotics application, including T1: 0%; T2: 25%; T3: 50%; T4: 75%; T5: 100% fertilization rates. The water-saving techniques promoted for rice were taken as suggested by Chen et al. (2018). A periodic drying and re-flooding irrigation were allowed to dry the field for few days before re-flooding, without stressing the plants. The irrigation water depth in the pot was kept to 3 cm and the irrigation interval was scheduled at 5-7 days, cracking of soil, depends on the weather conditions. The probiotic, Bacillus subtilis, were applied to each cultivation pot 3 to 5 days after the nitrogen fertilizer application.

2.2 Agronomic Parameters Assessment

Growth parameters, plant height and tiller number, were measured once every seven days from planting to harvest. Plant height was measured from the base to the tip of the highest leaf. Moreover, the Soil Plant Analysis Development – 502Plus (SPAD, Minolta Camera Co., Osaka, Japan) chlorophyll meter was used to determine the relative amount of chlorophyll once every seven days from planting to harvest. SPAD 502Plus measures the difference between the transmittance of a red (650 nm) and an infrared (940 nm) light through the leaf. It could provide a rapid and non-destructive approach that enables users to measure chlorophyll content in the leaf of rice plant. Furthermore, at harvest, yield components (total tillers number, effective and ineffective tillers, yield per plant) were obtained from each cultivation pot.

3. RESULTS AND DISCUSSION

3.1 Plant Height

The plant height of fertilized groups was begin to increase gradually after 50% tillering stage, but the non-fertilized treatment did not grown well until DAT 35 (Figure 1). Moreover, in the fertilized treatments, the plant height was significantly higher than non-fertilized treatment from the 50% tillering stage to the heading stage, but there was no significant difference within fertilized group. After heading stage, 100% and 75% groups was higher than 50%, 25% and non-fertilized treatments. However, there was no significant difference between 100% and 75%. It indicated that there is no significant difference among the 25, 50 75 and 100% fertilization rate during the vegetative period, except the non-fertilized treatment. But, during the heading stage, the performance of plant height were separated into two groups, 75 - 100% and 0 - 50%.

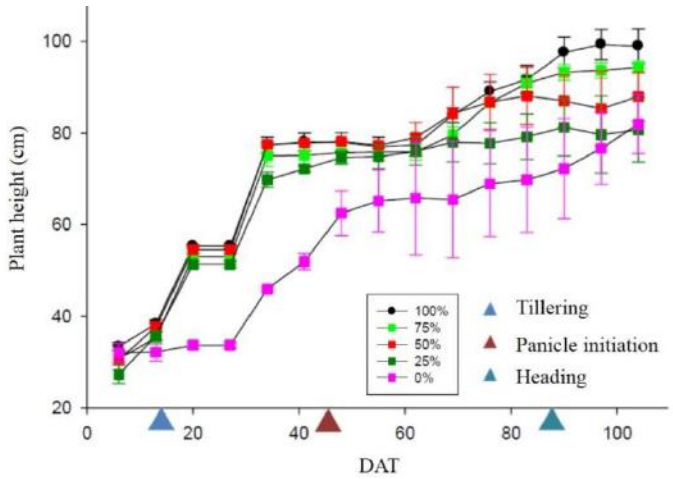


Figure 1. The trend of plant height

3.2 Tiller Number

In the performance of tillers number, there were obviously three groups separated from five fertilizer application rate treatments since 50% tillering stage (Figure 2). The 100% and 75% fertilization rate had the highest number of tillers. And, the trend of tillers development was similar between 100 and 75% fertilizer application rates. The tiller status at panicle initiation stage were shown as Figure 3

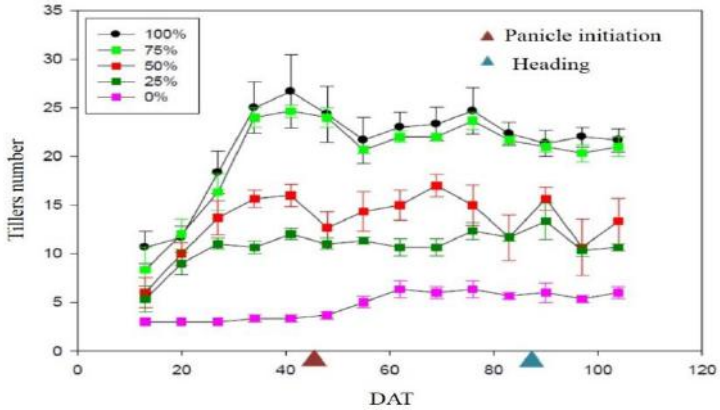


Figure 2. The trend of tiller numbers

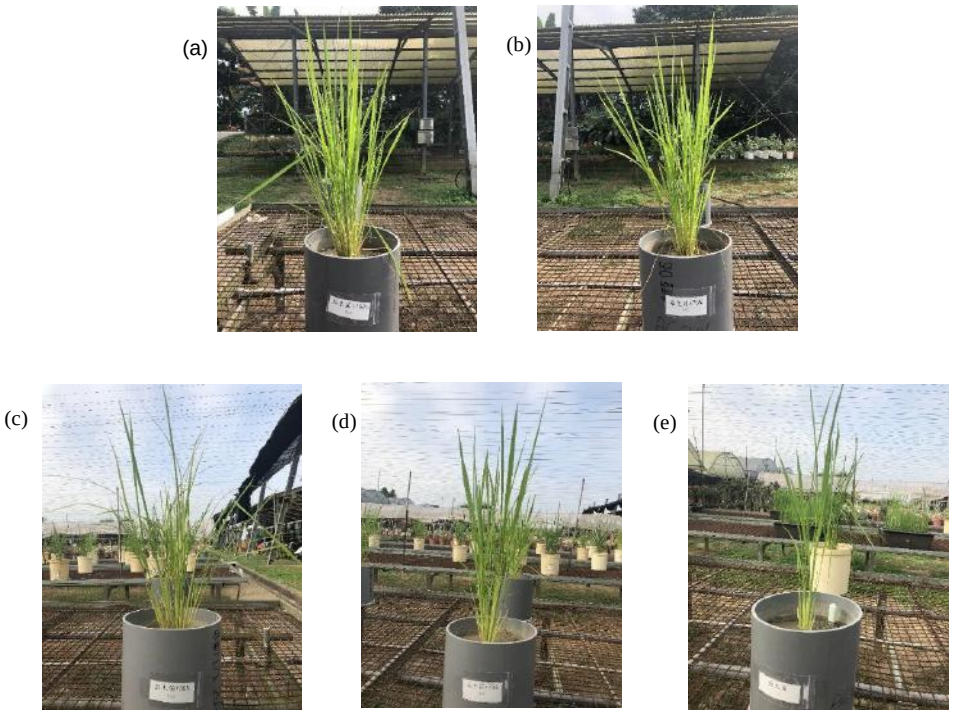


Figure 3. The tiller status at panicle initiation stage in different fertilizer application rates. 100%,(b) 75%,(c)50%,(d) 25%,(e) 0%

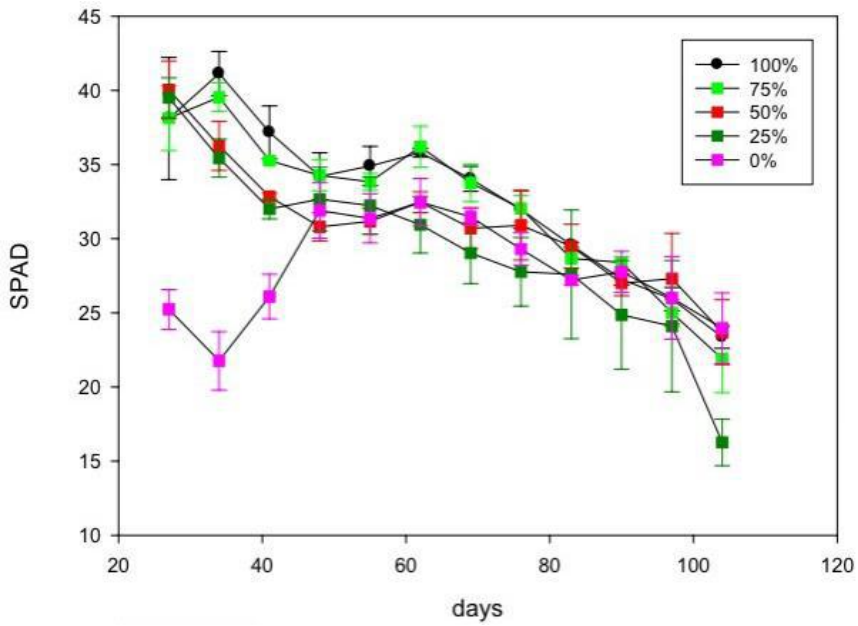


Figure 4. The trend of SPAD value

4. SPAD VALUE

SPAD 502Plus chlorophyll meter was used to determine the relative amount of chlorophyll in the leaf of rice plant once every seven days from planting to harvest by rapid and non-destructive approach. For the rice plants, a higher SPAD value, the healthier plant. As shown in Figure 4, the trend of SPAD values were decreasing indicated that the plant nitrogen concentration were diluted (dry weight increment) or decreasing (nitrogen deficient), according to the reports of Peng et al. (1995), Stroppiana et al.(2009), and Li et al. (2014). After fertilizer application, the SPAD values improved from 45 DAT to 60 DAT. Moreover, from planting date to the heading stage, the SPAD values of 100 and 75% fertilizer application rates were higher than of 0, 25, and 50%.

5. Yield Components At Harvest Stage

The experimental results of yield components at harvest stage were shown as Figure 5 – 7. The performance of tillers number values were separated about three groups, 75 - 100%, 25 - 50% and 0% fertilizer application rates. Although, the variation degree of total tillers number at 50% fertilizer application rates was higher (Figure 5). Moreover, the ineffective tillers number among all fertilizer application rate treatments had no significant differences. However, the effective tillers number of 75% fertilizer application rates was higher than 0, 25, and 50%. But, it was insignificant difference as compared with the 100% fertilizer application rates (Figure 6). Furthermore, the yield of each plant of 75% fertilizer application rates was also higher than 50% treated group, and similar to 100% fertilizer application rates (Figure 7). 0% fertilizer application rates, adding probiotic without fertilizer application, had the lowest yield.

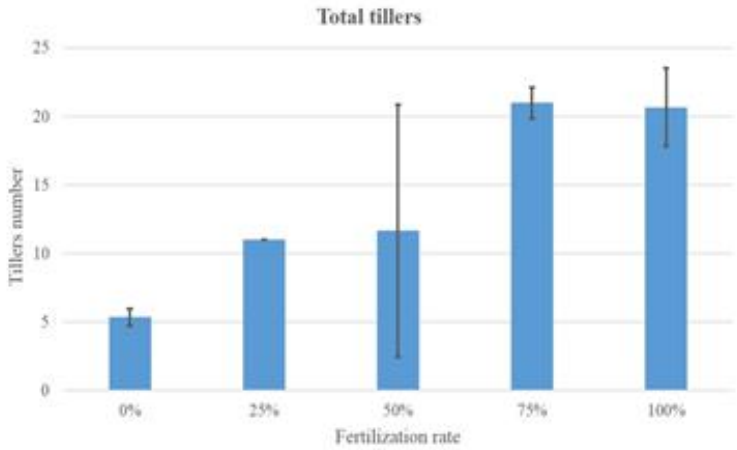


Figure 5 Total tillers at harvest stage

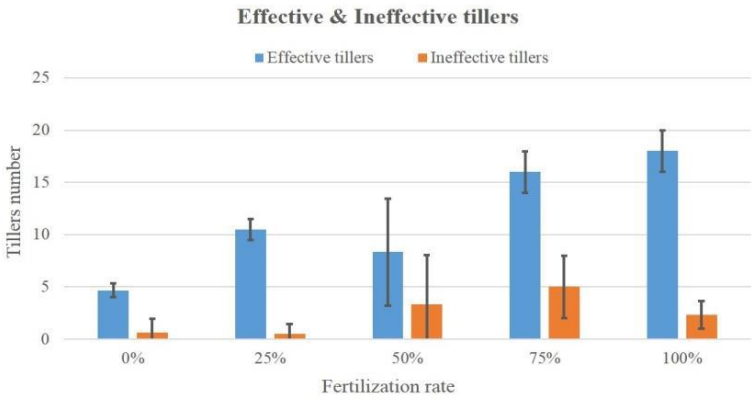


Figure 6 Effective and ineffective tillers at harvest stage

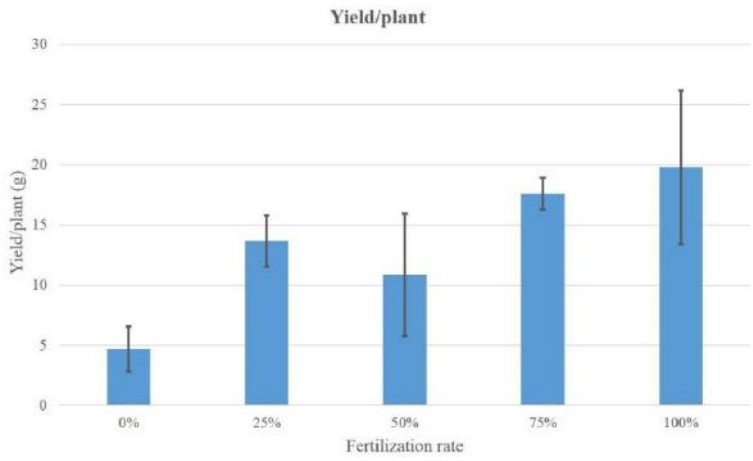


Figure 7 Yield of each plants at harvest stage

6. CONCLUSIONS

Due to the increasing water scarcity, irrigation water availability is decreasing, significantly impacts on agricultural production, especially in rice cultivation. Therefore, the main objective of this study is to assess the efficiency of the integrated AWD with probiotics application, by reducing the water and nitrogen fertilizer usage, and then providing an alternative approach to achieve eco-friendly production of paddy rice.

The study revealed that there's a threshold demand of nitrogen though incorporating with probiotics. In contrast, the compensation effect was not obvious in the plant height of non-fertilized group and the tiller numbers of 50 and 25% fertilization application rate. The reason might be the development of the probiotics were suppressed by the limited nitrogen source condition. Furthermore, the total tillers number and effective tillers number could still be sustained by 75% fertilization application rate similar to 100%, such that the final yield of each plants was the same as 100% treatment. The similar evidence also appeared in SPAD values of all fertilization application rates. In conclusion, against the increasing of water scarcity, the integrated AWD irrigation approach with probiotics application could reduce the water and nitrogen fertilizer usage without reducing quantitative traits, even in crop yields. For fertilizer application, it will reduce its rate by at least 25%. All in all, the integrated AWD irrigation approach with probiotics application is a promising option for rice producers to provide an alternative approach to achieve eco-friendly production of paddy rice.

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